
Fraudulent Transactions Impact

Predictive Intelligence in BFSI

Advanced Fraud Detection & Explainable AI System leveraging Machine Learning and LLMs for real-time security.



Financial Fraud Poses Critical Threats to BFSI Institutions

The Challenge

Traditional rule-based systems fail to detect sophisticated, evolving fraud patterns in modern banking.

The Solution

A hybrid approach combining **Machine Learning** for high-speed detection and **LLMs** for transparent decision-making.

Key Features

- Real-time risk scoring for every transaction.
- Explainable AI (XAI) for human-readable reasoning including AI Advisor Chatbot.
- Scalable API-first architecture for enterprise integration.



Strategic Goals of Predictive Transaction Intelligence

Predictive Accuracy

Utilizing advanced algorithms to distinguish legitimate from fraudulent patterns with high precision and minimal error.

Pattern Recognition

Analyzing historical transaction data to identify subtle anomalies and behavioral shifts that indicate potential fraud.

Explainable Insights

Delivering LLM-powered reasoning to help compliance officers understand the specific factors behind every flagged transaction.



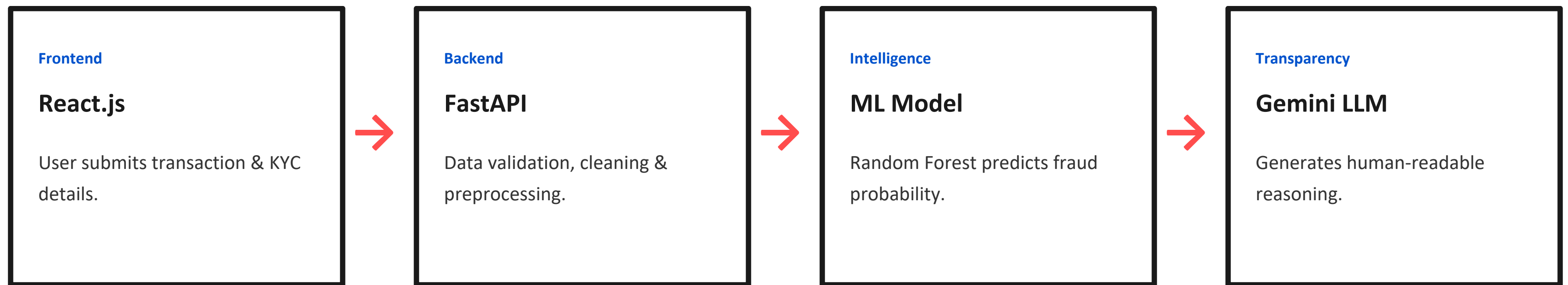
Operational Efficiency: Reducing false positives for a seamless customer experience.



Modern Tech Stack for Scalable Fraud Detection

Layer	Technology Stack
Frontend	React.js, JavaScript, Framer Motion (UI Animations)
Backend API	FastAPI (High-performance Python), Axios
ML & Data	Scikit-learn (RandomForest), Pandas, NumPy
AI & LLM	Google Gemini AP (Explainability), SQLite
Visualization	Chart.js (Frontend), Matplotlib & Seaborn (EDA)
DevOps	Joblib (Model Storage), python-dotenv, Postman

System Architecture & Data Flow Pipeline



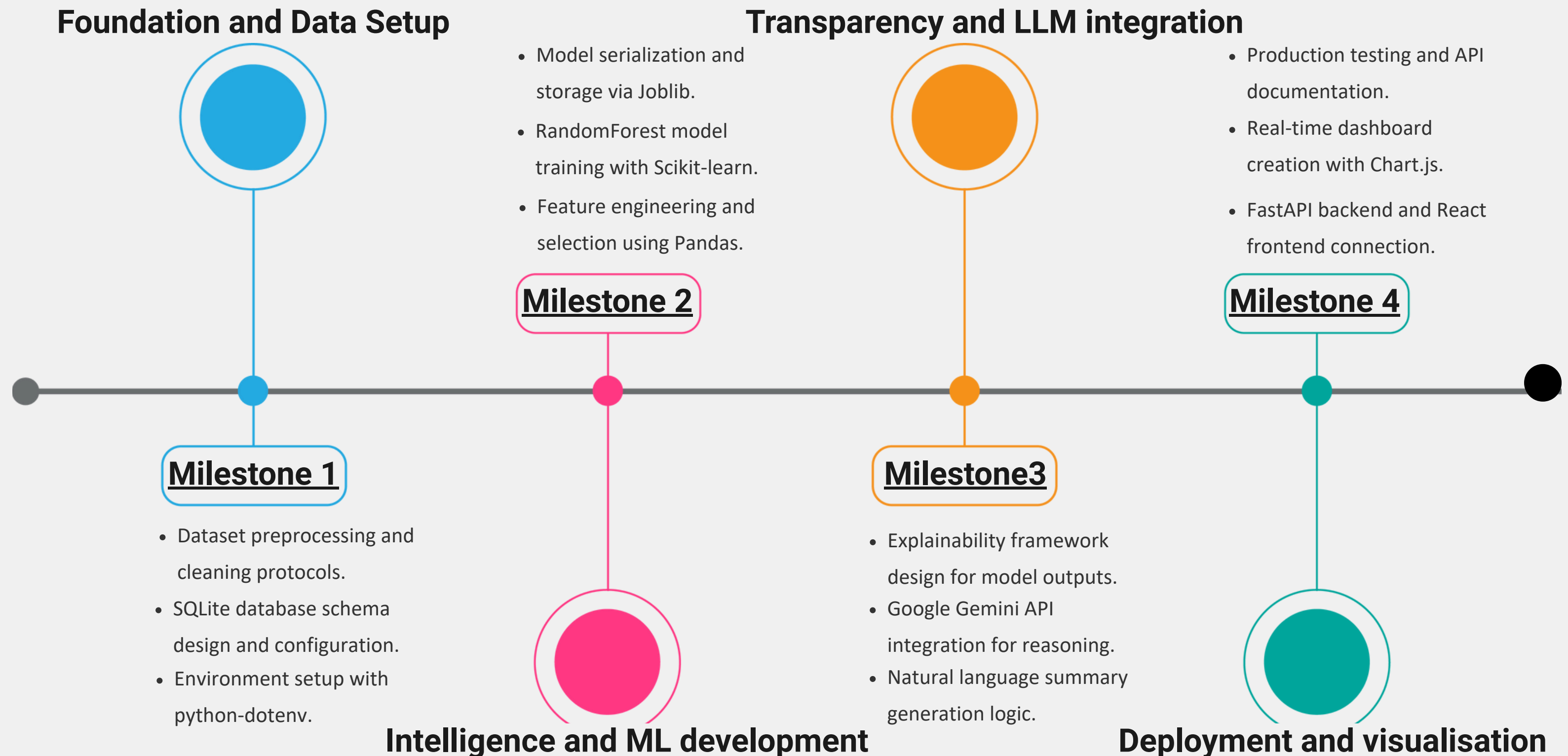
01 Data Input: Users submit transaction details via a responsive React interface.

03 Intelligence: Scikit-learn's RandomForest predicts fraud probability scores.

02 Processing: FastAPI handles validation and preprocessing using Pandas and NumPy.

04 Transparency: Gemini API interprets the model's output into natural language insights.

Project Milestones & Development Roadmap



RESTful API Endpoints for System Integration

POST/api/verify

Validates KYC data and ensures user authenticity before transaction processing.

POST/api/predict

Analyzes transaction details to return a real-time fraud probability score.

GET/api/explain

Retrieves LLM-generated reasoning behind specific model predictions for transparency.

GET/api/history

Accesses historical verification and prediction logs for auditing and compliance.



Project Output & Visual Insights

AI-Powered BFSI Intelligence Platform

Real-time fraud detection, risk prediction, and financial intelligence using advanced Machine Learning. Secure your transactions with the power of AI.

[Login](#)

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Smarter Decisions for Modern Banking

- Real-time fraud and risk detection
- Scalable AI models for large transaction volumes
- Secure, compliant, and audit-ready platform

Built for Trust. Powered by Intelligence.

- ✓ Detect anomalies before they become losses
- ✓ Reduce manual review effort with smart alerts
- ✓ Gain explainable insights, not just predictions

Why Choose BFSI Intelligence?



AI Fraud Detection

Detect anomalies and suspicious patterns in real-time using our trained Random Forest models.



Transaction Risk Scoring

Every transaction gets a risk score (High, Medium, Low) to help you prioritize reviews.



Smart Alerts System

Get instant notifications for high-risk activities so your team can act immediately.



Secure Data Upload

Support for CSV, Excel, and AI-powered Parsing for PDF/Image documents.



Real-time Insights

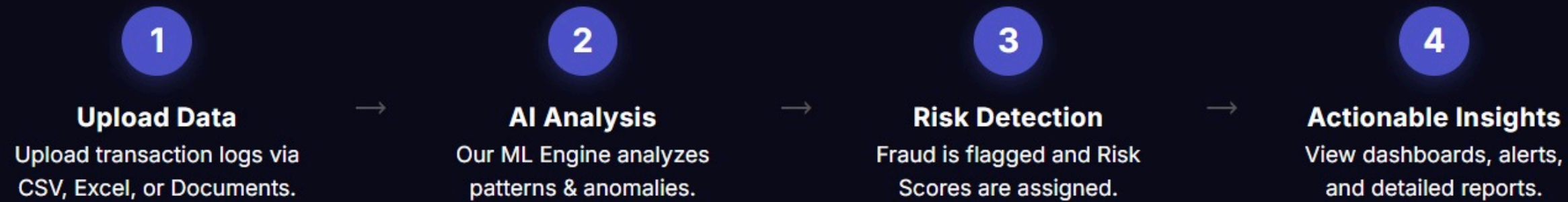
Visual dashboards with live metrics, trends, and actionable intelligence.



Predictive Analytics

AI suggests model improvements and strategies to stay ahead of new fraud techniques.

How It Works



From Data to Decisions — Seamlessly

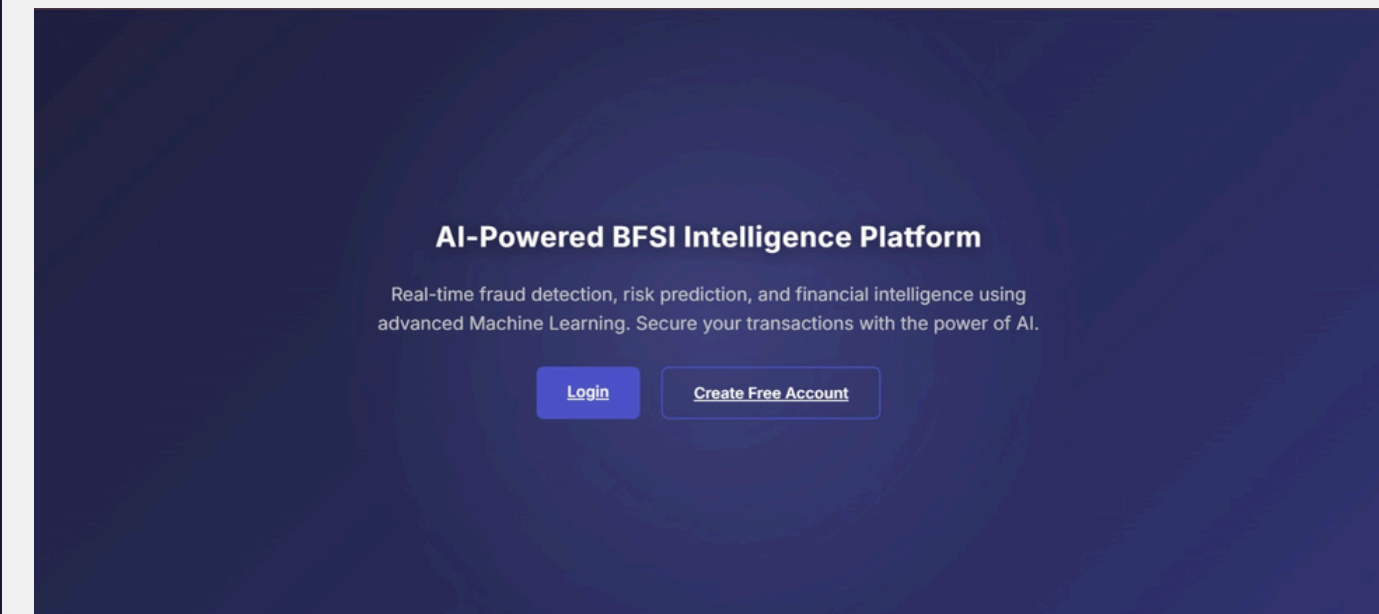
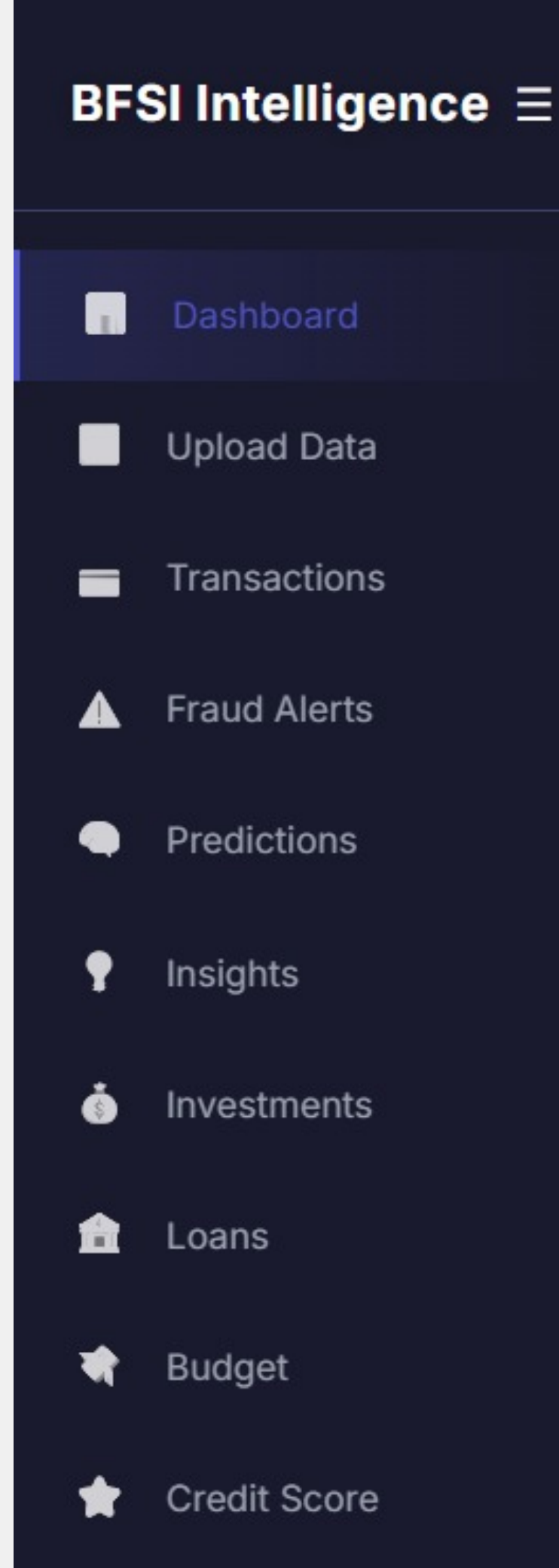
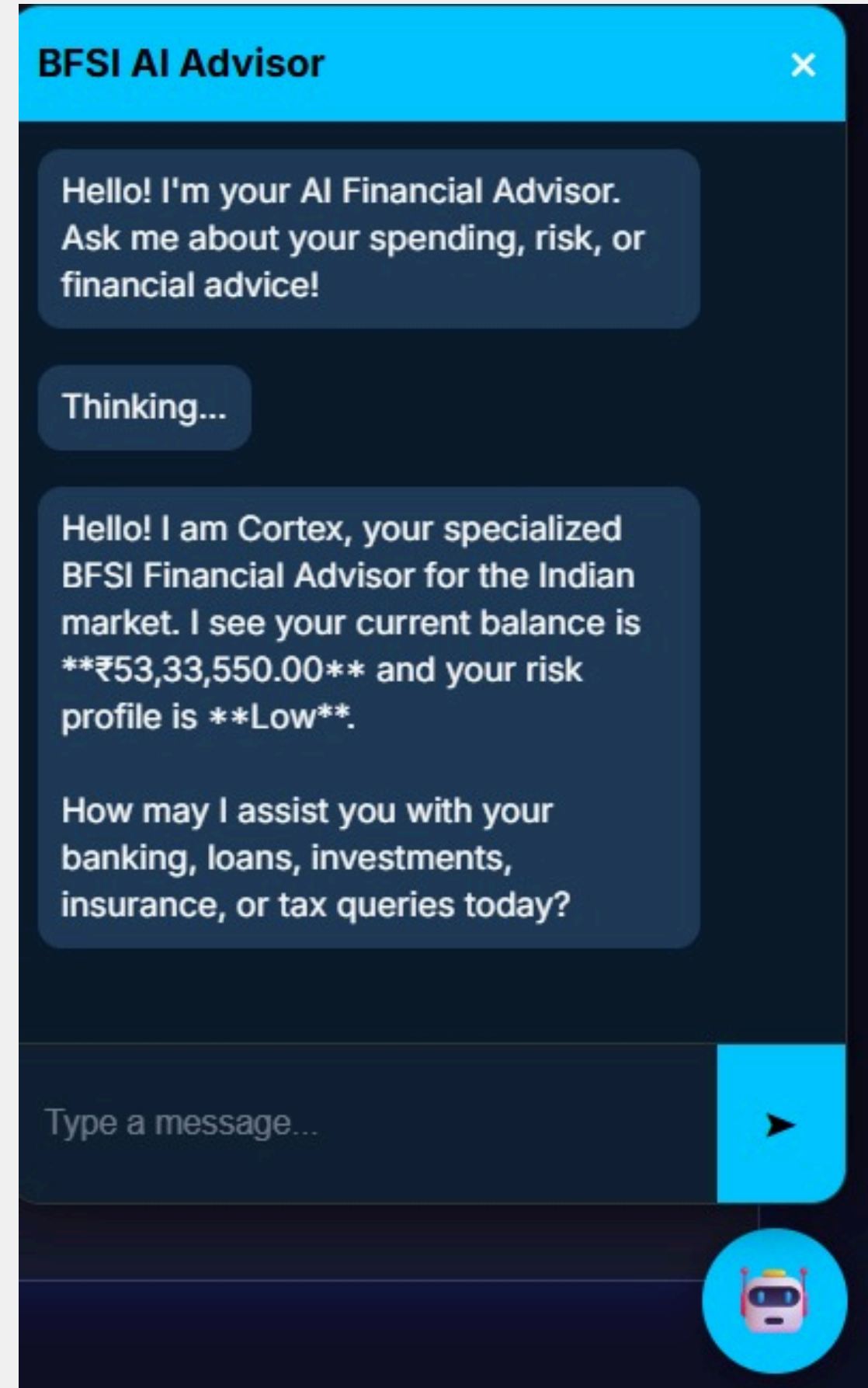
Our platform transforms raw financial data into actionable intelligence through a streamlined, automated workflow. No complex setup. No manual intervention. Upload your data, let AI do the heavy lifting, and act instantly on insights that matter.

Dashboard UI

Monitor transactions, detect suspicious behavior, and track risk scores — all from a single, intuitive dashboard designed for speed and clarity.

- Live transaction monitoring
- Intelligent fraud alerts
- Trust & risk score visualization
- Clear trends and performance metrics





Future Enhancements for Enterprise Readiness

Real-Time Stream Processing

Integrating live data pipelines for instant fraud prevention and automated alerts across banking channels.

Edge AI Deployment

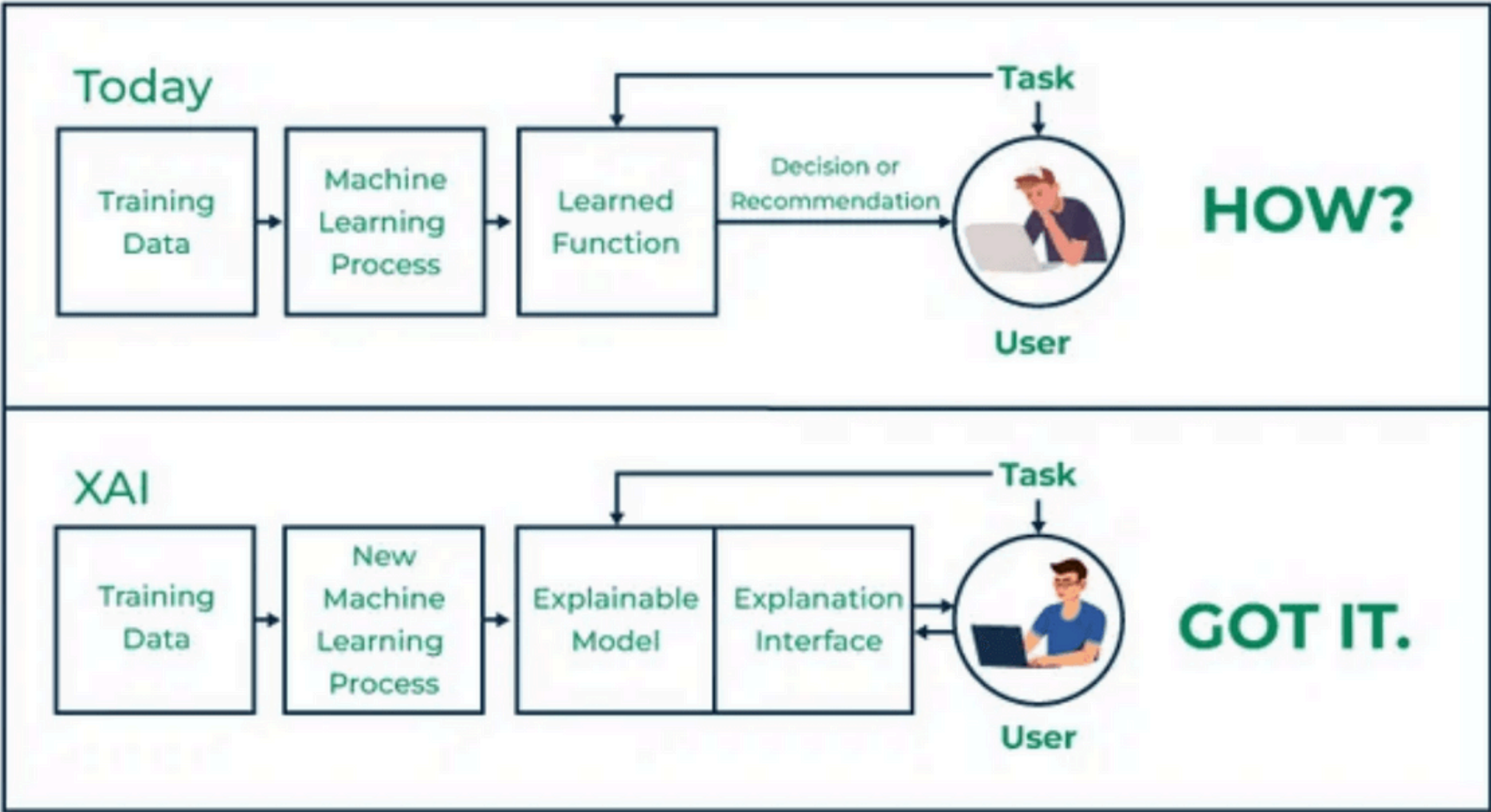
Enabling on-device prediction for faster and more secure transaction validation without cloud latency.

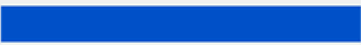
Blockchain Integration

Maintaining tamper-proof KYC and transaction verification records for enhanced auditing and trust.

Federated Learning

Implementing collaborative, privacy-preserving model training across multiple financial institutions.





Thank You

Any Questions?

Project Mentor

Kapil Panwar

Project

Predictive Transaction Intelligence using for BFSI